

Chapter 18

Global Quantum Strategies



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Abstract Many countries have created national quantum strategies in order to centralize or coordinate quantum research, innovation, commercialization, and talent development in their countries. This chapter comparatively analyzes four national quantum strategies from Canada, Denmark, the Netherlands, and the UK. Each country embraces its own model for quantum strategy, with some favoring large, generative investments, while others pursue more decentralized or additive investments. The overall financial commitments may seem large in the aggregate but constitute a modest per annum spending. Countries that have yet to create their own national strategy can learn from the diverse models for building quantum ecosystems.

18.1 Introduction

Experts debate the precise timing for a commercial “breakthrough” in quantum technologies, yet there is widespread agreement that long-term investments are required, particularly for quantum computers with useful real-world advantages and applications. These investments are being made not just in research and development, but in the wider ecosystems that will support a new quantum economy. For this broad set of tasks, governments have been playing an increasing role through the deployment of national funding schemes, programs, and initiatives. To coordinate these efforts, over a dozen countries have gone further by developing national quantum *strategies*.

Quantum strategies are generally more involved than quantum programs or initiatives, in that they are multi-stakeholder, cross-sectoral, and consider wider social implications. They create both a vision for a quantum-enabled society and concrete plans or funding mechanisms for bringing sectors together to achieve

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this vision. According to the OECD, national science, technology, and innovation strategies generally incorporate the following characteristics:

- They express the government’s vision on how the scientific field can contribute to the country’s economy and society.
- They create public investment priorities and how to create structures to enable them.
- They help rally private and/or public actors around a common goal and facilitate coordination to enable a common vision [1].

Switzerland, while recently launching a Swiss National Quantum Initiative (NQI), has yet to create or deploy a national quantum strategy. However, several of its peer countries have. The aim of this chapter is to describe in detail the quantum strategies of four countries—Canada, Denmark, the Netherlands, and the United Kingdom—and then analytically compare their strategies and provide preliminary recommendations for Switzerland or other countries that may be considering creating their own strategy. While the two global leaders in quantum technologies, the USA and China, are also important, it is more difficult to draw useful comparisons given their large size and very different models—especially in the case of China, where public information is lacking and national autonomy is a primary aim.

These four countries were selected because they have developed extensive quantum strategies, because they are considered among the leading nations in quantum technologies, because they prioritize international engagement, and because the size of their populations and quantum programs provide considerable diversity as a basis of comparison.

18.2 National Quantum Strategies

18.2.1 *Canada*

Canada’s National Quantum Strategy (short: “Canadian Strategy”), created by the Canadian Ministry of Innovation, Science, and Industry, debuted in January 2023, although spending had begun in 2021. The Canadian Strategy consists of three interlocking missions: to make Canada “a world leader” in quantum technologies, to focus on privacy and cybersecurity, and to become early adopters through both government and industry. The three core pillars to achieve these missions are research, talent, and commercialization, with CAD 360 million (CHF 228 million) in targeted public funding planned over seven years starting in 2021 [2].¹

¹ All currencies were converted to CHF using prevailing market rates at the time of authorship. Euros and USD were not converted.

Program implementation through funding streams comes from a wide array of preexisting organizations and initiatives. Major funders for research and development of quantum technologies include the Canada First Research Excellence Fund (CFREF) and Natural Sciences and Engineering Research Council (NSERC). Through NSERC, a new “Alliance Quantum” funding scheme is devoting around CAD 130 million (CHF 82 million) over seven years to allow for greater research and development collaborations—with CAD 100 million (CHF 63 million) targeted for domestic collaborations and CAD 30 million (CHF 19 million) for international collaborations. The first round of international grants in 2024 provided up to CAD 300,000 (CHF 190,000) per project over 1–3 years [2, 3].

The focus of the Canadian Strategy leans heavily toward the computing sector, with the goal to amplify Canada’s strong position in quantum hardware and software. Focus areas include hybrid computing, quantum simulators, and use cases in fields such as digital security, banking, or advanced manufacturing. On the commercialization side, innovation challenge programs are issuing calls for advancements such as quantum computing as a service, smaller dilution fridges, or computing prototypes [2]. There is also support promised for the quantum sensing sector, which follows an analogous strategy of investing in research, backing prototypes, supporting common standards, and using the government as an early adopter for technologies [2].

Canada benefited from significant early private and research investment, which helped to create a pioneering and robust startup ecosystem in quantum technologies. The country ranks second globally in the number of quantum startups (after the USA), with 28 in total. D-Wave launched the first commercial quantum-based computer, a quantum annealer, in 2011. Overall, estimated private funding in the last two decades into Canadian quantum technologies has totaled CAD 1.3 billion (CHF 825 million), close to the CAD 1.4 billion (CHF 888 million) in public investment, according to McKinsey. In fact, total investment in quantum startups in Canada from 2001 to 2023 has exceeded all investments in EU startups in the same time frame [4]. Incubators have played a large role, especially the Creative Destruction Lab from the University of Toronto, which has helped launch more than 50 quantum-related companies. The government has pursued commercialization through large grants and investments directly in companies, as well as challenge programs and regional innovation clusters such as Quantum City in Calgary or DistriQ in Sherbrooke [2, 5]. The Canadian Strategy is thus able to build its funding for commercialization on the shoulders of a very robust quantum private industry.

Given these strengths, Canada is highly optimistic about the scale of quantum technologies in its long-term future economy. The Canadian National Research Council estimated that by 2045 the national quantum industry would be valued at CAD 139 billion (CHF 88 billion) and constitute 3% of the country’s GDP [2, 5]. If the quantum economy achieves this scale, then significant investments in talent and workforce development are needed—an area of action for the Canadian Strategy. Much of the national initiative for talent development comes from individual university programs, such as a new Bachelor’s in Quantum Science at Université de Sherbrooke or a new Master’s at the University of Waterloo. The Canadian

Strategy is also providing CAD 40 million (CHF 25 million) to Mitacs for internship and professional development training in quantum [6]. The government is likewise involved in broader science outreach programs to increase diversity in quantum [2].

Quantum security is a major pillar of the Canadian Strategy. Canada hopes to identify vulnerable communications and assets, work with researchers on post-quantum cryptography solutions, and also potentially build a quantum secure network [7]. In these activities, the government sees itself as enabling research and development, being an early adopter of critical technologies, and also promoting international collaboration and innovation exchange [2].

On the international front, Canada's association into Horizon Europe in 2024 has opened up new international funding and collaboration possibilities, although the full impact is still unfolding and will take time to assess [8]. Bilateral agreements with UK and US funding agencies also cover scientific exchange more broadly. The "Alliance Quantum" grants will specifically target international collaboration in quantum with funding at CAD 30 million (CHF 19 million) over seven years.

18.2.2 *Denmark*

Denmark's National Strategy for Quantum Technology (short: "Danish Strategy") was published in two parts by the Ministry of Higher Education and Science in June and September 2023. On a four-year timeline through 2027, the government intends to commit DKK 1 billion (CHF 126 million) for quantum research and innovation, and an additional DKK 200 million (CHF 25 million) for commercialization, security, and international collaboration [9].

The Danish Strategy funding programs are mainly administered via Innovation Fund Denmark. The fund is expected to shift its focus over time toward demonstration and scale-up technologies as the ecosystem matures. In order to advise the direction of programs and allocation of funds, a National Forum for Quantum Technology has been established [10]. The strongest academic research clusters to support this ecosystem are located at the University of Copenhagen, Technical University of Denmark, and Aarhus University.

Danish government expenditure is supplemented by significant private-sector funding. Most notably, the Novo Nordisk Foundation has established a Quantum Computing Program at the University of Copenhagen, pledging DKK 1.5 billion (CHF 189 million) in 2022 for a 12-year program to build fault-tolerant quantum hardware and algorithms to solve life-science problems [10]. To leverage private-sector computing expertise, part of the Novo Nordisk funding has been used to establish an IBM Quantum Hub at the University of Copenhagen [9]. Moreover, KPMG established its Global Quantum Hub in Denmark, collaborating with leading universities [11]. More recently, in February 2024, Novo Nordisk Foundation also pledged DKK 150 million (CHF 18.9 million) to establish a Copenhagen Center for Biomedical Quantum Sensing [12].

In commercialization of quantum technologies, two concrete measures stand out in the Danish Strategy. Denmark pledges to establish a Quantum House Denmark with approximately DKK 16 million (CHF 2 million) yearly funding. This physical hub will provide lab and office space, business development guidance, and networks spanning universities, startups, funders, and corporations. Closely connected will be a national test center, funded through Novo Nordisk. The second concrete measure is a new Quantum Fund administered by Denmark's Export and Investment Fund—although exact funding will be dependent on private co-investment and leftover funds from other initiatives [9].

In contrast to some other countries, Denmark only has a very modest startup ecosystem in quantum technologies. Sparrow Quantum, producer of single-photon chips, and Kvantify, a hybrid software builder, are among the most notable Danish players. Two other Danish quantum companies were recently acquired by international players: QDevil by Quantum Machines and NKT Photonics by Hamamatsu.

As part of the EU, Denmark benefits significantly from EU funding instruments such as Horizon Europe and the EU Quantum Flagship. From 2014 to 2020, Denmark received around DKK 500 million (CHF 63 million) from the EU framework programs for quantum projects [10]. Denmark is establishing strong security guardrails in navigating the international environment, particularly with non-EU actors, and prioritizes collaboration with NATO and other strong allies [9]. In 2023, Denmark was selected as the site of a new NATO Center for Quantum Technologies, hosted at the Niels Bohr Institute in Copenhagen [13].

Denmark's key internationalization initiative will be a new International Quantum Hub funded with DKK 10 million (CHF 1.25 million) yearly through 2027. It is intended as the primary entry point for international quantum collaborators to reach Danish authorities and the quantum ecosystem. The hub will likewise represent a coherent tone and Danish viewpoint on international and security policy questions related to quantum technologies [9].

18.2.3 Netherlands

The Netherlands' National Agenda for Quantum Technology (short: "Dutch Strategy") was published in September 2019. Uniquely among the countries covered, the Dutch Strategy created a single prominent institution, Quantum Delta Netherlands (QDNL), to execute the Dutch quantum strategy. QDNL was proposed as a "world-leading center and hub for quantum technology" analogous to Silicon Valley [14]. QDNL describes itself as an ecosystem, developing its own initiatives while also consolidating the naming of preexisting quantum centers under the umbrella of "Delta hubs" [15].

QDNL is uniquely positioned as a distinct nonprofit agency that is mandated to execute the government's strategy. To this end, the Dutch government in 2020 invested an initial EUR 23 million, followed in 2021 by an additional EUR 615 million over seven years, in QDNL to coordinate the Dutch quantum ecosystem.

This model has created a uniquely powerful and centralized institution that is nonetheless able to operate distinctly from government ministries [16].

QDNL has developed three specific technology catalyst programs in quantum computing and simulation, quantum networking, and quantum sensing. Additionally, they have four main action lines devoted to building quantum research, talent, societal impact, and the wider ecosystem. Much of the presence of QDNL in the ecosystem comes through initiatives, events, and programs; however, they also have built a physical site directly through the House of Quantum in Delft. This center was designed to support research, startup space, and facilities and programs that inspire interaction and collaboration among quantum players [17]. Other House of Quantum locations are planned, with the goal of creating an interconnected “national campus” for quantum technology development [18]. The development of a Dutch quantum national strategy has benefitted from the geographic density of quantum expertise. The three main preexisting quantum research centers (QuTech in Delft, QuSoft in Amsterdam, and QT/e in Eindhoven) are within roughly two hours travel distance [14].

Public investment in quantum technologies, research, and infrastructures far exceeds private investment in the Netherlands. As of 2024, the Netherlands has pledged a total of around EUR 1 billion in public investments. This dwarfs the roughly EUR 40 million in private investments made in Dutch quantum technologies, with around a dozen quantum companies active in the Netherlands [4].

Dutch international activities in quantum have largely focused on EU cooperation. In addition to being a core participant in Horizon Europe and EU Quantum Flagship programs, QDNL has set up a trilateral research program with Germany and France. The Dutch contribution to the trilateral program of EUR 62 million will fund joint centers of excellence, as well as strategic research and development projects through a trilateral call [19].

18.2.4 United Kingdom

The UK published its National Quantum Strategy (short: “UK Strategy”) in March 2023. They had already committed GBP 1 billion (CHF 1.1 billion) in public funding since 2014 through the National Quantum Technologies Programme, but starting in 2023 have pledged another GBP 2.5 billion (CHF 2.8 billion) in the following 10 years [20]. The UK Strategy programs from 2024 to 2034 will be split into two five-year phases. The first phase includes new research hubs and centers, outcome-focused innovation programs, industry-led innovation, accelerator programs, training and talent development, international collaboration, and infrastructure investments [20].

Like many other countries, the UK is motivated by international competition. “We are in a global race,” the UK Strategy states. “The UK was an early mover, but other countries are accelerating their own efforts” [20]. The UK Strategy’s mission outlines several core performance metrics to achieve by 2034: for example,

to maintain its top three national position in quality and impact of quantum publications; to fund 1,000 new postgraduate students; and to develop five new bilateral funding arrangements. On the innovation front, they aim to attract 15% of total global private equity investment to the UK, have a 15% share of the global quantum technology market, and motivate 75% of UK companies to prepare for quantum computing. Most of these figures represent meaningful leaps over current figures [20].

The UK has been an early mover in quantum research and innovation. Since 2014, the UK has funded a network of four Quantum Research Hubs at universities and institutions, such as the Quantum Metrology Lab at the National Physical Laboratory and the National Quantum Computing Centre (NQCC), focused on building a scalable device. This preexisting Quantum Program touts the distributed network of research hubs across the country, which also includes the Fraunhofer Centre for Applied Photonics in Glasgow, Scotland. In 2024, the UK government announced five new hubs for quantum technology development with an additional investment of GBP 160 million (CHF 178 million) [21].

The UK's early strength in quantum innovation is evidenced by around 50 quantum startups that have emerged (with government assistance) since 2014—the most in Europe. These companies have attracted a combined GBP 425 million (CHF 474 million) in private investment [20]. At least 160 companies are active in the UK quantum sector, which ranks second globally in private equity investment after the USA [20, 22]. The UK Strategy aims to leverage its GBP 2.5 billion (CHF 2.8 billion) in public investment to attract an additional GBP 1 billion (CHF 1.11 billion) in private funding by 2033 [20]. The UK Strategy also hopes to attract further quantum businesses to the UK, such as with the return of PsiQuantum or the IBM collaboration on a National Centre for Digital Innovation [20, 23].

Aside from challenges general to quantum technology deployment, specific UK challenges include demand for quantum skilled labor, higher salaries, especially in the USA, and competition from larger countries and multinational corporations—all of which can erode the UK's strong early position in the “quantum race” [20]. To compete for talent, the UK has special workforce development provisions. They aim to double the number of PhDs in quantum or supporting technologies in the 2024–2034 period to over 1,000, with an initial 2-year investment of GBP 25 million (CHF 28 million). They also will expand a STEM outreach program in schools and push for enhanced visa pathways for foreign talent [20, 24].

The UK Strategy emphasizes the application of quantum technologies in a few key sectors to strengthen their overall economy and society, in particular sustainability, security/defense, life science, and data analytics/AI. The SparQ program run through the NQCC will explore these and other early applications. Defense, in particular, is seen as a useful early adopter and purchaser of quantum technologies, providing government support for emerging technologies [20]. More broadly, the NQCC, in addition to serving as a research and development hub, will also explore critical applications, standards, and regulations and act as a first purchaser for prototype platforms [20].

The UK has made notable advances in international partnerships. The government signed its first major joint statement for bilateral cooperation with the USA in 2021. The UK Strategy's goal is to develop this into a "comprehensive program of bilateral arrangements." Since 2021, additional quantum MoUs or partnerships have been announced with several countries, including Australia, Canada, Denmark, South Korea, and Germany [20].

18.3 Comparative Analysis

Given the different timescales, goals, and country population sizes, it is very difficult to make a strict quantitative comparison between the quantum strategies of Canada, Denmark, the Netherlands, and the UK. One can, however, roughly compare the pledged public investment on a per person, per annum (pp/pa) basis. On this metric, Canada's public quantum investment in its national strategy is modest, at less than CHF 1 pp/pa. The UK and Netherlands each have pledged slightly over CHF 4 pp/pa in their strategies, while Denmark has pledged around CHF 6.4 pp/pa. Thus, while the headline spending numbers for the various quantum strategies are designed to impress, the amount of taxpayer funds going toward supporting the emergence of this specialized technology is relatively modest.

Qualitative comparisons between the different countries' strategies reveal both fundamental differences in philosophy, but also similar strategies to shepherd quantum technologies to commercial viability. Canada and the UK are both notable for their very strong, early private sector investments in quantum research and innovation. The impressive array of quantum startups and established quantum companies in the two countries has been backed by an ecosystem of research translation, accelerators, supply chains, and investors, which combine to give Canada and the UK a strategic advantage. Their quantum strategies thus seek to build on these existing strengths. Denmark and the Netherlands, in contrast, are confronted with the task of building up a quantum commercial ecosystem from a less robust prior basis.

These different backgrounds inform the direction of investments and the level of centralization. Canada arguably has the "lightest touch" in terms of overall government coordination, as its effort can be seen mainly as a funding boost for regional hubs and bottom-up research and development clusters. On the opposite end of the spectrum is the Netherlands, which has created a new centralized organization, QDNL, that provides a single platform for all strategic quantum funding and activities. The UK sits in between these poles, investing heavily in regional clusters but also working to elevate the National Quantum Computing Centre into a robust institution for promoting its strategic priorities across sectors.

The level of government involvement can be seen in the relative balance of public vs. private investments. While all quantum strategies hope to create a robust commercial quantum sector, the UK and Canada aim to use their investments explicitly to attract sizable private sector funds. According to McKinsey, the UK

has spent or pledged CHF 3.9 billion in public investment, compared to CHF 1.4 billion in private monies. In Canada, it is CHF 890 million public investment for CHF 820 million in private money [4]. In contrast, the level of prior private investment in quantum companies is comparatively small in the Netherlands and Denmark. In Denmark specifically, government investment is exceeded by a single huge corporate foundation—Novo Nordisk—which exerts major influence to steer quantum technologies toward use cases in biopharma and the life sciences. The question remains if the government or foundation investment can eventually yield a self-sustaining commercial industry, or if it instead leads to continued large-scale public subsidies. The German case may be the starkest example of this imbalance, with nearly CHF 5 billion in public investments spent or pledged, compared to around CHF 100 million in private investment [4].

Given these very large public investments, many countries feel that they are in a competitive landscape against other governments. The Dutch and UK strategies explicitly mention this; the Dutch Strategy argues that “urgent action [is] required” since “other countries are not standing still” [14]. Because of the government-driven nature of national quantum strategies, there is often significant weight placed on defense applications and government procurement of quantum technologies as an early boost for quantum commercialization. This was prominent in most quantum strategy documents.

The international strategies in the national quantum strategies are often among the least concrete sections. Overall, it is politically and bureaucratically challenging to direct significant funds to foreign research and development. Nonetheless, different strategies emerge. The UK, Canada, and the Netherlands have all engaged in bilateral (or trilateral) calls to build up specific relationships with other countries based on perceived mutual strengths and geographic proximity. This is especially valuable for countries outside EU programs. The Netherlands and Denmark also benefit from being core EU member states and are well integrated into the EU Quantum Flagship. Canada and the UK are also set to benefit from EU framework funding. Canada has also pledged significant funds for international research collaborations through “Alliance Quantum” grants. In a different vein, Denmark has pledged to establish an International Hub that serves as a single “open door” for international collaborators.

18.4 Recommendations

It is not feasible to make any specific policy recommendations for Switzerland or other countries without quantum strategies, based solely on this survey of several other countries’ national quantum strategies. There is no single strategy for success, as each country has sought to identify and capitalize on its particular strengths.

In the Swiss case, the role of the Swiss National Quantum Initiative (NQI) will likely entail strengthening the core values and priorities of the Swiss quantum ecosystem, as identified by the Swiss Quantum Commission to include:

- Support for bottom-up, curiosity-driven research and innovation, rather than top-down directives
- Development of shared infrastructure and platforms
- Avoidance of direct subsidies, in favor of market processes and private competition
- Support education, workforce, and quantum literacy efforts
- International engagement
- Long-term, continued support necessary to reach quantum technology breakthroughs and commercial viability [25]

These priorities suggest that an overarching Swiss Quantum Strategy, if developed, may entail a “lighter touch” system of support for ongoing research and innovation, similar to the Canadian model, rather than a more centralized program as in the Dutch case. Nonetheless, given the perceived competition among national governments for resources and talent, and Switzerland’s modest size, the country may also benefit from a more coordinated “public face”—especially for the public and international collaborators—that can help give visibility to Swiss excellence and build support for collaborations and public investments. As in Denmark and the Netherlands, the geographic density of Switzerland also lends itself to greater coordination of clusters, as opposed to dispersed regional hubs in countries like Canada. Other countries will need to respond to their specific conditions, such as the level of centrality in an organization and geographic density when designing their strategies.

In the Swiss case, reliance on bottom-up excellence in research, education, and innovation means that a national strategy will likely play a coordinating rather than a generative role in the development of a quantum ecosystem. Preexisting initiatives such as new master’s programs in quantum engineering and bachelor’s programs at multiple universities of applied sciences are already creating a growing workforce, and Switzerland can build on its strength as a world-leading talent factory. One main area where a Swiss Strategy might play a more “generative” role is in stimulating common infrastructure, perhaps even reaching an ambitious scale such as the UK’s National Quantum Computing Centre.

The UK provides a similar model to Switzerland: it is a European country with a strong foundation of research that has previously been excluded from core EU funding schemes and thus has been pushed to collaborate on a more bilateral basis internationally. The UK strategy of developing several bilateral agreements has been mirrored in early Swiss efforts, such as the Joint Statement of Cooperation in Quantum Technologies with the United States. The Swiss government has also signed agreements to promote broader advanced technologies, sometimes including quantum, with several other countries. As Swiss research and innovation thrive on international collaboration, it is essential to create programs and platforms that invite the outside world to work with Switzerland on quantum. This might include new funding schemes, an international hub, or programs and initiatives using preexisting international platforms such as Swissnex.

18.5 Conclusion

As of early 2024, governments worldwide have announced a combined USD 42 billion in public investments in quantum technologies, which includes USD 10 billion in new funding announced in 2023 [4]. Many of these countries are explicitly steering this money to create a broader quantum ecosystems, backed by national strategies.

Time will tell if these public investments can sustain themselves, or whether government impatience or shifting priorities lead to a “quantum winter” in terms of public commitment. The Swiss ecosystem has sustained itself thus far largely on bottom-up energy, an excellent education and innovation pipeline, and long-standing infrastructures. This suggests that the current Swiss quantum ecosystem is more resilient, since it is less dependent on a single national program or initiative.

Nonetheless, given the competitive landscape as countries struggle to make large quantum breakthroughs, Switzerland will want to consider carefully whether a national quantum strategy can secure its future continued leading position and relevance in the emerging global quantum economy. The four peer countries studied here—all of whom are also strong international collaboration partners for Swiss quantum actors—provide lessons on how to structure such a potential strategy.

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